Tear here

If you prefer that your surname and first names be treated as confidential, you can conceal it by folding and sealing this part of the page.

Fold here

Important: Fill in you surname, first names, and signature in the space provided.

SURNAME (please print) [redacted]

FIRST NAMES [redacted]

SIGNATURE [redacted]

Telephone number where you can be reached	
During examination peri[redacted]	
After examin[redacted]	

University of Pretoria

Additional answer books used must be placed in the back of the first book.

Number of books handed in: ....

Number of this book: ....

Registration number
[redacted]

8

DEGREE OR DIPLOMA COURSE (eg. Bsc. 1) BEng

MODULE (eg. Chemistry 110) COS132

NUMBER OF PAPER (eg. Paper 2) Semester test 1

DATE OF TEST / EXAMINATION 16/04/2024

In column 1 candidates must list the questions the answered in the order in which they were answered.

Column 1	Column 2	
Number of question	Marks awarded by examiners	
	Int.	Ext.
1	15	
2	8	
3	7	
4	8½	
5	7	
6	14	
Points awarded: Total	59.5	

FOR EXAMINERS:

1 Examiners must enter in column 2 the marks obtained for each question.

99% outstanding!

2 **INITIALS OF EXAMINERS:**

Internal:

1. ....

2. ....

3. ....

External:

1. ....

2. ....

3. ....

Q1.) (a)

(i) False ✓

(ii) True ✓

(iii) True ✓

(iv) False ✓

(v) False ✓

(vi) True ✓

(vii) False ✓

15

(b)

(i) C ✓

(ii) B ✓

(iii) C ✗

(iv) B ✗

(v) D ✓

(vi) D ✓

~~(c)~~

(c)

(i) ✗

(ii) Parentheses ✓

(iii) iomanip ✓

(iv) precision ✓

(v) initialised ✓

Q2.) (a) 15 ✓

(b) 12 ✓

(c) 1 3.2 ✓

~~(d)~~

(d) 0
1

8

~~Q4.)~~

Q3.) Syntax error on line 4: There is ~~no~~ a missing semi-colon at the end of the line

1

Syntax error on line 5: There is a missing double quote at the end of the string literal

1

Syntax error on line 10: The second operator in the cout statement is not an insertion operator.

1

Semantic error on line 9: The expression will divide by zero.

2

~~Semantic error on line 10: There There is a string literal~~

Semantic error on line 8: ~~There is A~~ narrowing implicit will occur ~~with~~ when "z" is initialised.

Q4.) (d) Identification Decomposition Abstraction Algorithm / Application

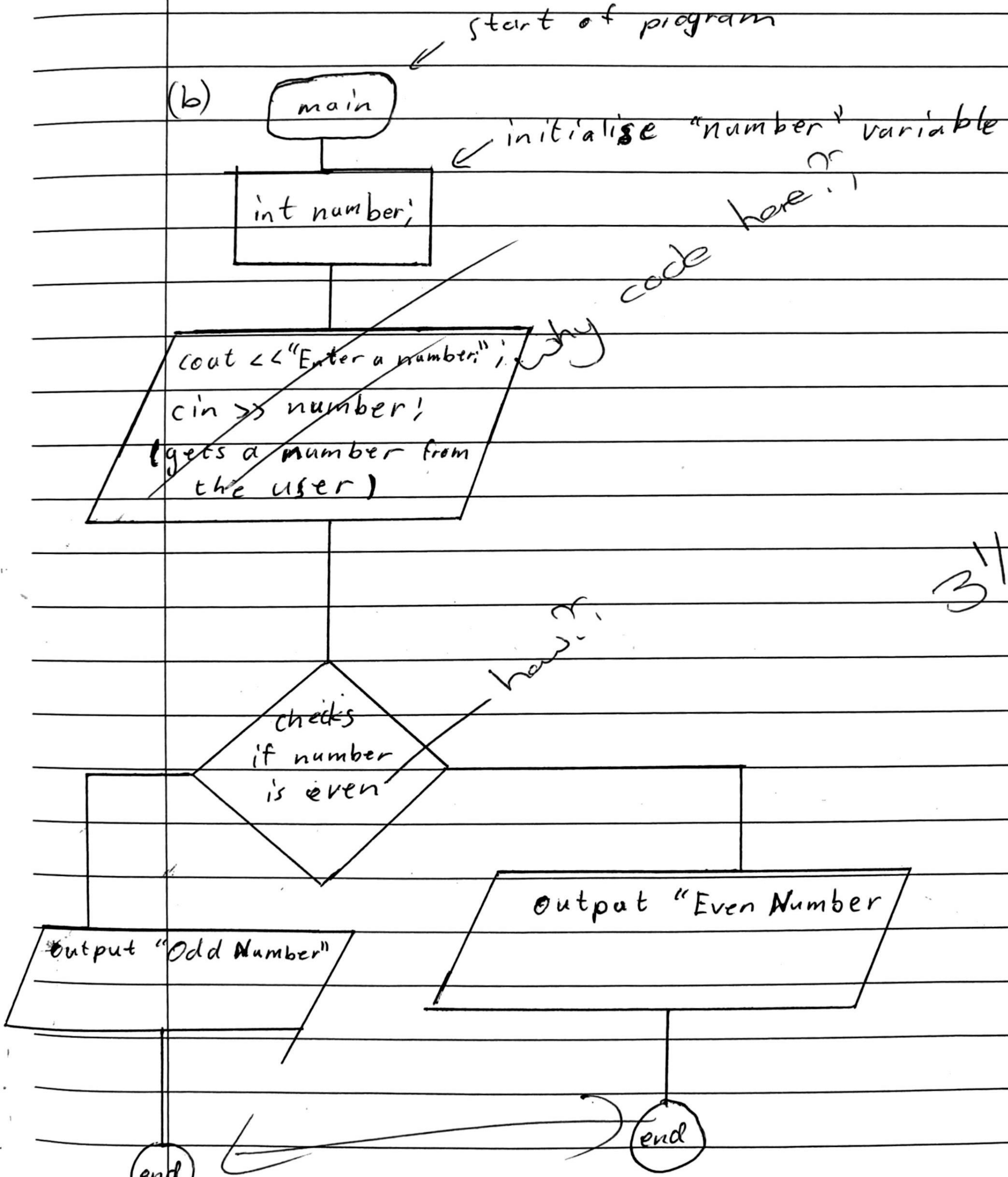
Identify the problem: Create ~~an~~ steps to follow to organise the food courses and ensure a satisfying dining experience

Decomposition: Need to invite guests, ensure enough seating is arranged, have all meals reddy to be served ~~at the right time~~, and ~~an~~ a good ammount of time should be planned for in between courses.

Abstraction: Call guests well before the date. Get ingredients for meal courses. Cook meals and make deserts. Serve guests efficiently and pleasantly.

Algorithm: I should use my phone to contact my guests, I should prepare all the courses before the guests arrive and organise them in the right order, I should plan the time between courses and finalise each course before serving. 5

(b)



3½.

Q5.)(a)

~~#include <iostream>~~

#include <iostream>

(b)

~~const float PI;~~

The most appropriate method would be to define π(PI) as a ~~g~~ global constant ~~float~~ double variable.

1

(c) ~~co:~~

~~const float PI = 3.141592654~~

const double PI = 3.141592654;

(d) cin >> radius >> height;

(e) ~~static_cast <float>~~

static_cast <double>(PI * radius * radius * height)

1

Q6.)(~~A~~)(a)(i)

float areaOfTriangle(float base, float height){
    return (0.5 * base * height);
}

(ii)

~~float areaOfCircle(float radius) {~~

~~float~~

~~float areaOfCircle(float radius) {~~

~~return (3.141592654 * radius * radius);~~

float areaOfCircle(float radius) {
    return (PI * radius * radius)
}

(b)(i)

~~Calculator:~~

~~Advanced_Maths.o:~~

Q6 1) (a)

(i)

```
float areaOfTriangle(float base, float height) {
    float area = multiply(0.5, multiply(base, height));
    return area;
}
```

(ii)

```
Float areaOfCircle(float radius) {
    float area = multiply(PI, multiply(radius, radius));
    return area;
}
```

(b)(i)

~~Advanced_Maths.o : Advanced_Maths.cpp Constants.h~~

```
Advanced_Maths.o : Advanced_Maths.cpp Advanced_Maths.h Constants.h
[tab] g++ -c -std=c++98 Advanced_Maths.cpp

Maths.o : Maths.cpp Maths.h
[tab] g++ -c -std=c++98 Maths.cpp
```

(ii)

```
Main: Calculator.o Advanced_Maths.o Maths.o
[tab] g++ -std=c++98 Calculator.o -o Main
```

(iii) make run

(iv) ~~./Main~~ ~~make~~ ./Main or make

$\frac{14}{15}$